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GROUP NAME: FINAL SIX

Data Warehousing Preparation and Business Intelligence Queries

INSIGHT/DOCUMENTATION REPORT

This project helped our group gain a deeper understanding of how databases go beyond simply storing information and actually play a big role in producing meaningful business insights. Through designing both a relational schema and a star schema, we were able to clearly see the difference between transactional systems and analytical systems. We learned that while a normalized relational schema is ideal for everyday operations like inserting and updating data, it becomes more complex and slower to use when it comes to generating reports because of the multiple joins required.

When we designed the star schema, the concept of Business Intelligence (BI) became much clearer for us. We saw how the fact_sales table acts as the central point for analysis, while dimension tables such as dim_date, dim_customer, and dim_product provide the necessary context for understanding the data. This structure made querying much simpler and improved performance, especially when working with larger datasets.

Another important takeaway for our group was writing BI queries that reflect real business questions. Tasks like calculating monthly revenue, identifying top-selling products, and grouping customers based on their purchasing behavior helped us understand how SQL can support decision-making. We also gained more confidence in using aggregate functions, grouping, filtering, and CTEs to uncover patterns and insights from the data.

One of the challenges we faced was maintaining data consistency across multiple tables, especially when dealing with joins and conditions such as cancelled orders. It required careful checking to ensure that the results were accurate and meaningful. We also found that creating useful customer segments and metrics like lifetime value required not just technical skills, but also analytical thinking.

In Conclusion, I this project strengthened our skills in database design, SQL querying, and data analysis. More importantly, it helped us understand how proper data structuring can make reporting more efficient and support better business decisions. As a group, we now feel more prepared to apply these concepts to real-world analytics and more complex database systems in the future.